

Molecular pathways for cancer chemoprevention by dietary phytochemicals.

Interest in dietary phytochemicals for potential cancer chemoprevention has increased substantially. Screening dietary compounds for chemopreventive activity however, requires a systematic and wide-ranging approach to encompass the complexity of carcinogenesis. We present some of the molecular pathways that underpin the broad biological processes involved in carcinogenesis. Oxidative stress, inflammation, and the evasion of apoptosis are important biological mechanisms by which carcinogenesis occurs. Subsequently, antioxidant, anti-inflammatory, and pro-apoptotic activity represent important activities for preventing, suppressing, or reversing the development of carcinogenesis. Ultimately, these mechanisms of action may provide a useful basis for screening novel phytochemicals for chemopreventive activity. In this review, we identify the important molecular processes that may be targeted in routine screenings of dietary phytochemicals to ultimately select the most effective potential candidates for cancer chemoprevention.